

## ORIGINAL ARTICLE

## Left Atrial Remodeling and Functional Maladaptations in Obesity: An Invasive Pressure-Volume Analysis

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**BACKGROUND:** Obesity is a major risk factor for atrial fibrillation and heart failure. We aimed to characterize left atrial (LA) structural and functional changes across body mass index (BMI) strata in patients with atrial fibrillation referred for catheter ablation.

**METHODS:** We studied 1040 consecutive patients (67% male; mean age, 62 years; 71% in sinus rhythm). Participants were stratified by BMI (<25.0, 25–29.9, 30–34.9, and ≥35 kg/m<sup>2</sup>). Comprehensive echocardiographic assessment of LA structure and function was integrated with invasive LA pressure measurements obtained via transseptal access during the ablation procedure. Associations across BMI strata were evaluated using regression analyses.

**RESULTS:** Patients with higher BMI had larger estimated total blood and plasma volumes along with higher LA volumes and pressures, greater cyclic wall stress, and a rightward shift of the estimated LA pressure-volume relationship. Conversely, estimated operant LA stiffness did not differ across BMI strata. BMI-related differences in LA volume were attenuated by body surface area indexing but preserved with height-based indexing. Conventional volumetric indices of LA phasic function were similar across BMI strata, whereas available LA strain indices (n=107) were lower with greater adiposity, consistent with worse LA phasic function. Compared with patients without heart failure, those with heart failure exhibited more pronounced structural, functional, and hemodynamic atrial remodeling, consistent with a more advanced atrial myopathy phenotype.

**CONCLUSIONS:** In patients undergoing atrial fibrillation ablation, higher BMI is associated with a predominantly load-related pattern of LA remodeling, characterized by larger LA volume and pressures, greater cyclic wall stress, and a rightward shift of the estimated pressure-volume relationship. BMI-associated early atrial remodeling may precede overt atrial dysfunction, potentially contributing to the link between excess adiposity, atrial fibrillation, and heart failure with preserved ejection fraction.

**Key Words:** atrial fibrillation ■ echocardiography ■ heart failure ■ middle aged ■ phenotype

Obesity has become a major global health concern, with a rapidly increasing prevalence and a substantial clinical, social, and economic burden.<sup>1</sup> It is a well-established risk factor for several cardiovascular disorders, including atrial fibrillation (AF),<sup>2</sup> and heart failure with preserved ejection fraction (HFpEF).<sup>3</sup>

Increasing body mass index (BMI) is typically associated with cardiac remodeling, typically left ventricular hypertrophy,<sup>4</sup> yet its impact on the left atrium (LA) remains less clearly defined. In particular, it is uncertain to what extent obesity promotes intrinsic atrial structural disease<sup>5</sup> versus predominantly inducing hemodynamic,

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### WHAT IS NEW?

- We provide a large invasive human characterization of left atrial remodeling across the spectrum of body mass index in patients undergoing atrial fibrillation ablation. By integrating echocardiographic and invasive hemodynamic data, we show that obesity is associated with progressively larger left atrial volumes, higher atrial pressures, greater cyclic wall stress, and a rightward shift of the left atrial pressure-volume relationship, without a corresponding increase in estimated operant atrial stiffness. These findings suggest a predominantly load-related stage of atrial remodeling that may precede overt atrial myopathy.

### WHAT ARE THE CLINICAL IMPLICATIONS?

- These findings provide mechanistic insight into the association between obesity, atrial fibrillation, and heart failure. They suggest that, in patients with obesity, left atrial enlargement may initially reflect increased hemodynamic loading rather than advanced intrinsic atrial dysfunction. They also highlight that indexing left atrial volume to body surface area may attenuate or obscure obesity-related differences, whereas height-based indexing better preserves them. These observations are hypothesis-generating but may provide a biologically plausible rationale for future studies evaluating obesity- and congestion-targeted interventions.

### Nonstandard Abbreviations and Acronyms

|                          |                                                |
|--------------------------|------------------------------------------------|
| <b>AF</b>                | atrial fibrillation                            |
| <b>BMI</b>               | body mass index                                |
| <b>BSA</b>               | body surface area                              |
| <b>HF</b>                | heart failure                                  |
| <b>HFpEF</b>             | heart failure with preserved ejection fraction |
| <b>LA</b>                | left atrium                                    |
| <b>LAEF</b>              | left atrial emptying fraction                  |
| <b>LAV</b>               | left atrial volume                             |
| <b>LAV<sub>max</sub></b> | maximal left atrial volume                     |
| <b>LV</b>                | left ventricular                               |
| <b>LVEF</b>              | left ventricular ejection fraction             |
| <b>PV</b>                | pressure-volume                                |

volume-driven LA enlargement,<sup>6</sup> which is the most commonly used marker of atrial remodeling.<sup>7</sup> Several mechanisms linked with obesity, such as higher estimated circulating and plasma volume, increased cardiac output, and chronic volume loading, could lead to passive LA expansion without necessarily implying early atrial myopathy.<sup>8–10</sup> Consistently, while Mendelian randomization studies suggest a causal association between higher

BMI and AF,<sup>11</sup> the intermediate atrial phenotype through which obesity predisposes to AF remains incompletely understood.

Here, we aimed to characterize LA structural and functional changes associated with BMI strata by integrating morpho-volumetric, functional, and pressure-related measures. This multimodal approach sought to provide a more refined understanding of BMI-related LA remodeling and its potential relevance to AF and heart failure (HF) development in a large cohort of patients referred for catheter ablation of AF.

## METHODS

### Data Availability

Deidentified individual participant data underlying the findings reported in this study are not publicly available because of institutional and ethical restrictions but may be made available from the corresponding author on reasonable request, subject to approval by the relevant authorities.

### Study Protocol

The study prospectively included consecutive patients undergoing AF ablation at the Institute for Clinical and Experimental Medicine in Prague (CZ). This study protocol adhered to the principles of the Declaration of Helsinki and was approved by the local ethics committee. All patients signed informed consent for the procedures and for participation in this research study. The study followed the Strengthening the Reporting of Observational Studies in Epidemiology reporting guideline for cross-sectional studies.

At the time of hospital admission, patients underwent a standardized clinical assessment including medical history review, physical examination with anthropometric measurements, transthoracic echocardiography, ECG, and blood sampling. All investigations were performed in a postabsorptive state following an overnight fast. BMI was calculated as weight/height<sup>2</sup>. Body surface area (BSA) was calculated by using the Du Bois formula ( $BSA [m^2] = 0.007184 \times \text{height (cm)}^{0.725} \times \text{weight (kg)}^{0.425}$ ). Obesity was defined as a BMI  $\geq 30$  kg/m<sup>2</sup> according to the World Health Organization.<sup>12</sup> Estimated glomerular filtration rate was calculated using the Modification of Diet in Renal Disease equation. The CHA<sub>2</sub>DS<sub>2</sub>-VASc score, cardiovascular risk factors, and medication use were collected at admission from clinical records. Diabetes and hypertension were identified from medical records and ongoing specific treatment. HF was defined as a preexisting clinical diagnosis documented in the medical records and a left ventricular ejection fraction (LVEF)  $\leq 40\%$ . As the diagnosis of HF was not systematically adjudicated, the HF phenotype could not be reliably classified.

### Echocardiography and Invasive Measures

Echocardiographic examinations were performed by experienced operators using standard, commercially available ultrasound equipment (Vivid 9 and Aurora Vivid 95, GE Healthcare) during the index hospitalization. Cardiac structure and function were measured according to contemporary recommendations.<sup>13</sup> Mitral and tricuspid regurgitation were assessed



semiquantitatively and expressed in 5 grades (absent/trivial [0], mild [1], moderate [2], moderate-severe [3], and severe [4]). Right ventricular function was evaluated in the apical 4-chamber view using tricuspid annular systolic excursion. LA volumes were obtained using the disk summation method in the apical 4-chamber view.<sup>14</sup> Volumetric assessment included maximal LA volume ( $LAV_{max}$ ; frame before mitral valve opening), diastasis LAV before atrial contraction (frame before LA contraction, or at the time of the P wave on the ECG), and minimal LAV (frame before mitral valve closure). Diastasis LA volume was measured only in patients in sinus rhythm. Global and reservoir LA function were characterized by total LA emptying fraction (LAEF;  $[LAV_{max} - \text{minimal LAV}]/LAV_{max} \times 100$ ),<sup>15</sup> LA conduit function was characterized by passive LAEF ( $[LAV_{max} - \text{LAV before atrial contraction}]/LAV_{max} \times 100$ ), and contractile function was characterized by active LAEF ( $[\text{LAV before atrial contraction} - \text{minimal LAV}]/\text{LAV before atrial contraction} \times 100$ ).<sup>16</sup> Atrial function was also assessed by the LA function index, which normalizes function to stroke volume and is rhythm-independent.<sup>17</sup>  $LAV_{max}$  was expressed as absolute values and also as LAV indexed to BSA<sup>14</sup> and LAV indexed to height squared.<sup>18</sup>

LA speckle-tracking analyses were performed in a subgroup of patients using EchoPAC (GE Healthcare), in accordance with current recommendations.<sup>19</sup> LA endocardial borders were manually traced in apical 2- and 4-chamber views, with the zero-strain reference set at the R wave on the ECG (ventricular end-diastole). An automated tracking algorithm was applied, with manual adjustments performed as needed to optimize wall tracking. LA strain was quantified for the LA reservoir strain and LA conduit strain phases. In patients in sinus rhythm, booster pump strain (LA contractile strain) was measured at the onset of atrial contraction, and LA conduit strain was derived as the difference between reservoir strain and booster pump strain. Final strain values for each phase were calculated as the average of measurements obtained from the apical 2- and 4-chamber views and reported as absolute values to facilitate interpretation, with lower absolute values indicating worse phasic LA function.

The hemodynamic measurements were performed under mild sedation using the electrophysiology recording system (CardioLab, GE Healthcare). The LA pressure was measured during the current heart rhythm immediately after the LA was accessed using a 7.5-Fr braided transseptal introducer (Schwartz SL1, St. Jude Medical). LA pressure was measured at maximal LA volume (V wave), at minimal LA volume, and as the pressure-time integral over the cardiac cycle (mean LA pressure). A value of mean LA pressure  $\geq 15$  mmHg was defined as elevated.<sup>20</sup> LA pulse pressure was calculated as the difference between the LA pressure at maximal and minimal volume. Pressure readings reflected the moving average over 10 cardiac cycles. Pressure-volume (PV) slopes and Y-intercepts were derived from 2-point PV estimates using maximal and minimal pressure and volume values. Estimated operant LA diastolic stiffness was approximated as the slope of the linear regression relating minimal and maximal LA PV coordinates, as previously described.<sup>21–23</sup> Meridional LA wall stress was estimated using modified Laplace-based equations applied to maximal-to-minimal LA pressures and volumes, following established formulas,<sup>24</sup> and assuming a nominal atrial wall thickness of 0.2 cm.<sup>25</sup> LA wall stress change across the cardiac cycle was calculated as maximum – minimum LA wall stress.

Plasma volume was estimated using the formula  $(1 - \text{hematocrit}) (a + [b \text{ weight in kg}])$ , where  $a = 1530$  for men and 864 for women, and  $b = 41$  for men and 47.9 for women.<sup>26</sup> Total blood volume was estimated as plasma volume/(1 – hematocrit).

## Statistical Analysis

Participants were categorized into ordered BMI strata: normal (BMI  $< 25.0$  kg/m<sup>2</sup>), overweight (25.0–29.9 kg/m<sup>2</sup>), class I obesity (30.0–34.9 kg/m<sup>2</sup>), and class II or greater obesity ( $\geq 35.0$  kg/m<sup>2</sup>). Data are presented as mean  $\pm$  SD or median with interquartile range for continuous variables (according to distribution), and counts (n) and percentages (%) for categorical variables. Normality was assessed using the Shapiro-Wilk test. Overall differences across BMI strata were assessed using 1-way ANOVA or Kruskal-Wallis tests for continuous variables, according to data distribution, and the  $\chi^2$  test for categorical variables. For the estimated LA PV relationship, the Jonckheere-Terpstra test was used to assess an ordered shift across BMI strata. Linear regression analyses were used to assess associations between continuous variables (ie, BMI, estimated circulating volumes, LA structural and functional measures, or hemodynamic parameters). Multivariable linear regression models were used in selected analyses to assess whether these associations persisted after adjustment for clinically relevant potential confounders. All tests were 2-sided, and a  $P$  value  $< 0.05$  was considered statistically significant. Analyses were performed using JMP 19.0 statistical software (SAS Institute, Inc, Cary, NC).

## RESULTS

### Baseline Characteristics Across BMI Strata

A total of 1040 patients were included. Age was similar across BMI groups, ranging from  $61 \pm 12$  years in normal-weight patients to  $62 \pm 9$  years in those with BMI  $\geq 35$  kg/m<sup>2</sup> ( $P = 0.773$ ), whereas higher BMI was associated with a greater proportion of men (from 52% to 72%;  $P < 0.001$ ) and a higher burden of comorbidities, as reflected by a higher CHA<sub>2</sub>DS<sub>2</sub>-VASC score. Indeed, hypertension was more prevalent across BMI strata (from 39% to 85%;  $P < 0.001$ ), as was diabetes (from 4% to 33%;  $P < 0.001$ ) and a history of HF (from 12% to 26%;  $P = 0.019$ ). Medication use was also higher across BMI categories (ie, diuretics, from 15% to 56%; renin-angiotensin system blockers, from 31% to 81%; mineralocorticoid receptor antagonists, from 5% to 22%; oral anticoagulation, from 76% to 97%; all  $P < 0.001$ ).

Overall, 88% of patients underwent echocardiography either on the same day as the catheter ablation procedure or the day before, with a median interval of 1 day (interquartile range, 1–1). Across higher BMI strata, patients had larger left ventricular (LV) end-diastolic diameter (from  $49 \pm 5$  to  $56 \pm 7$  mm) and LV mass (from  $152 \pm 41$  to  $211 \pm 53$  g; both  $P < 0.001$ ), slightly lower LVEF (from  $57 \pm 6\%$  to  $54 \pm 9\%$ ;  $P = 0.021$ ), and higher estimated filling pressures (E/e' ratio:  $8 \pm 3$  to  $10 \pm 5$ ;  $P < 0.001$ ; Table 1). Patients with available LA strain

**Table 1. Baseline Characteristics According to Body Mass Index Strata**

|                                                      | BMI <25.0 kg/m <sup>2</sup><br>(N=179) | BMI 25.0–29.9 kg/m <sup>2</sup><br>(N=439) | BMI 30.0–34.9 kg/m <sup>2</sup><br>(N=298) | BMI ≥35.0 kg/m <sup>2</sup><br>(N=124) | P value |
|------------------------------------------------------|----------------------------------------|--------------------------------------------|--------------------------------------------|----------------------------------------|---------|
| Body mass index, kg/m <sup>2</sup>                   | 23.1±1.6                               | 27.5±1.4                                   | 32.2±1.4                                   | 38.1±2.9                               | NA      |
| Clinical characteristics                             |                                        |                                            |                                            |                                        |         |
| Age, y                                               | 61±12                                  | 61±10                                      | 62±10                                      | 62±9                                   | 0.773   |
| Males, n (%)                                         | 93 (52)                                | 309 (70)                                   | 214 (72)                                   | 80 (65)                                | <0.001  |
| Body surface area, m <sup>2</sup>                    | 1.8±0.2                                | 2.0±0.2                                    | 2.2±0.2                                    | 2.3±0.2                                | <0.001  |
| CHA <sub>2</sub> DS <sub>2</sub> -VASc score, points | 1.7±1.4                                | 1.8±1.4                                    | 2.0±1.3                                    | 2.4±1.3                                | <0.001  |
| Hypertension, n (%)                                  | 66 (39)                                | 265 (63)                                   | 220 (76)                                   | 101 (85)                               | <0.001  |
| Diabetes, n (%)                                      | 7 (4)                                  | 39 (9)                                     | 37 (13)                                    | 39 (33)                                | <0.001  |
| History of heart failure, n (%)                      | 21 (12)                                | 69 (17)                                    | 44 (15)                                    | 31 (26)                                | 0.019   |
| Atrial fibrillation, n (%)                           |                                        |                                            |                                            |                                        |         |
| Paroxysmal                                           | 140 (78)                               | 266 (61)                                   | 191 (64)                                   | 78 (63)                                | <0.001  |
| Nonparoxysmal                                        | 39 (22)                                | 173 (39)                                   | 107 (36)                                   | 46 (37)                                |         |
| Previous AF ablation, n (%)                          |                                        |                                            |                                            |                                        |         |
| 0                                                    | 134 (75)                               | 328 (74)                                   | 220 (74)                                   | 100 (81)                               | 0.561   |
| 1                                                    | 39 (22)                                | 86 (20)                                    | 59 (20)                                    | 19 (15)                                |         |
| ≥2                                                   | 6 (3)                                  | 25 (6)                                     | 19 (6)                                     | 5 (4)                                  |         |
| Systolic blood pressure, mm Hg                       | 124±18                                 | 128±19                                     | 130±19                                     | 131±21                                 | 0.005   |
| Diastolic blood pressure, mm Hg                      | 74±11                                  | 74±11                                      | 74±11                                      | 74±11                                  | 0.186   |
| Heart rate, beats per minute                         | 71±22                                  | 70±20                                      | 70±19                                      | 72±18                                  | 0.702   |
| Baseline rhythm, n (%)                               |                                        |                                            |                                            |                                        |         |
| Sinus rhythm                                         | 127 (71)                               | 315 (72)                                   | 202 (68)                                   | 81 (65)                                | 0.448   |
| AF                                                   | 52 (29)                                | 124 (28)                                   | 96 (32)                                    | 43 (35)                                |         |
| Estimated total blood volume, mL                     | 4331±539                               | 5016±472                                   | 5677±478                                   | 6243±493                               | <0.001  |
| Estimated plasma volume, mL                          | 2576±289                               | 2914±309                                   | 3265±300                                   | 3616±342                               | <0.001  |
| Laboratory values                                    |                                        |                                            |                                            |                                        |         |
| Hemoglobin, g/L                                      | 138±14                                 | 144±15                                     | 146±15                                     | 143±15                                 | <0.001  |
| Hematocrit, %                                        | 40.3±3.6                               | 41.8±4.1                                   | 42.4±3.8                                   | 42.0±4.0                               | <0.001  |
| Sodium, mmol/L                                       | 139.2±2.2                              | 139.3±2.4                                  | 139.3±2.3                                  | 139.6±2.1                              | 0.507   |
| Potassium, mmol/L                                    | 4.3±0.4                                | 4.3±0.4                                    | 4.3±0.4                                    | 4.3±0.4                                | 0.796   |
| eGFR-MDRD, mL/min per 1.73 m <sup>2</sup>            | 75±18                                  | 72±18                                      | 72±19                                      | 71±22                                  | 0.378   |
| Echocardiography                                     |                                        |                                            |                                            |                                        |         |
| LV end-diastolic diameter, mm                        | 49±5                                   | 52±6                                       | 54±5                                       | 56±7                                   | <0.001  |
| LV end-diastolic diameter index, mm/m <sup>2</sup>   | 26±3                                   | 26±3                                       | 25±3                                       | 25±3                                   | <0.001  |
| LV ejection fraction, %                              | 57±6                                   | 56±6                                       | 56±7                                       | 54±9                                   | 0.021   |
| LV ejection fraction ≤40%, n (%)                     | 9 (5)                                  | 16 (4)                                     | 14 (5)                                     | 12 (10)                                | 0.048   |
| LV mass, g                                           | 152±41                                 | 181±46                                     | 199±45                                     | 211±53                                 | <0.001  |
| LV mass index, g/m <sup>2</sup>                      | 87±68                                  | 87±20                                      | 89±18                                      | 89±20                                  | 0.845   |
| E/E' ratio                                           | 8±3                                    | 8±3                                        | 9±3                                        | 10±5                                   | <0.001  |
| TAPSE, mm                                            | 23±6                                   | 22±5                                       | 23±5                                       | 22±5                                   | 0.904   |
| Mitral regurgitation grade, n (%)                    |                                        |                                            |                                            |                                        |         |
| 0–2                                                  | 168 (99)                               | 406 (98)                                   | 277 (98)                                   | 116 (99)                               | 0.385   |
| 3–4                                                  | 1 (1)                                  | 9 (2)                                      | 6 (2)                                      | 1 (1)                                  |         |
| Tricuspid regurgitation grade, n (%)                 |                                        |                                            |                                            |                                        |         |
| 0–2                                                  | 167 (99)                               | 419 (99)                                   | 278 (99)                                   | 109 (96)                               | 0.093   |
| 3–4                                                  | 1 (1)                                  | 2 (1)                                      | 2 (1)                                      | 4 (4)                                  |         |
| Treatment                                            |                                        |                                            |                                            |                                        |         |
| Any diuretic, n (%)                                  | 26 (15)                                | 123 (28)                                   | 91 (31)                                    | 70 (56)                                | <0.001  |

(Continued)

**Table 1. Continued**

|                              | BMI <25.0 kg/m <sup>2</sup><br>(N=179) | BMI 25.0–29.9 kg/m <sup>2</sup><br>(N=439) | BMI 30.0–34.9 kg/m <sup>2</sup><br>(N=298) | BMI ≥35.0 kg/m <sup>2</sup><br>(N=124) | P value |
|------------------------------|----------------------------------------|--------------------------------------------|--------------------------------------------|----------------------------------------|---------|
| ACE inhibitors or ARB, n (%) | 53 (31)                                | 226 (54)                                   | 187 (64)                                   | 96 (81)                                | <0.001  |
| MRA, n (%)                   | 9 (5)                                  | 35 (8)                                     | 14 (5)                                     | 27 (22)                                | <0.001  |
| β-Blocker, n (%)             | 99 (58)                                | 287 (69)                                   | 228 (78)                                   | 93 (78)                                | <0.001  |
| Antiarrhythmics, n (%)       | 116 (65)                               | 297 (68)                                   | 204 (69)                                   | 97 (78)                                | 0.085   |
| OAC, n (%)                   | 130 (76)                               | 349 (83)                                   | 258 (89)                                   | 115 (97)                               | <0.001  |

ACE indicates angiotensin-converting enzyme; AF, atrial fibrillation; ARB, angiotensin receptor blocker; eGFR, estimated glomerular filtration rate; LV, left ventricular; MDRD, Modification of Diet in Renal Disease; MRA, mineralocorticoid receptor antagonist; NA, not applicable; OAC, oral anticoagulant; and TAPSE, tricuspid annular plane systolic excursion.

measurements (n=107) were broadly comparable to those without strain data with respect to demographics, comorbidities, LA size, LA pressure, and other echocardiographic parameters (Table S1). Baseline clinical characteristics were also largely similar between patients in sinus rhythm and those with AF, although, as expected, AF was associated with larger LA volumes and more impaired LA function (Table S2).

### Blood and Plasma Volume

Estimated total blood and plasma volume were incrementally higher across BMI categories (from 4331±539 to 6243±493 mL, and from 2576±289 to 3616±342 mL, respectively), as were hemoglobin (from 138±14 to 146±15 g/L) and hematocrit (from 40.3±3.6% to 42.4±3.8%; all  $P<0.001$ ). In additional analyses adjusted for hypertension, diabetes, and HF/LVEF ≤40%, BMI was directly associated with estimated total blood volume ( $\beta$ , 137.03 mL per kg/m<sup>2</sup> [95% CI, 129.24–144.82];  $P<0.001$ ) and plasma volume ( $\beta$ , 73.26 mL per kg/m<sup>2</sup> [95% CI, 68.37–78.14];  $P<0.001$ ). To assess whether these findings were influenced by HF, in which formula-derived volume estimates may be less reliable, we repeated the analyses after excluding patients with HF and LVEF ≤40%, finding consistent positive associations (blood volume:  $\beta$ =132.81 mL per kg/m<sup>2</sup>; plasma volume:  $\beta$ =72.51 mL per kg/m<sup>2</sup>; both  $P<0.001$ ).

### LA Structure

Maximum LA volume was incrementally higher across BMI strata, ranging from 76±31 mL in normal-weight patients to 94±29 mL in those with BMI ≥35 kg/m<sup>2</sup> ( $P<0.001$ ; Figure S1A). Indexing to height<sup>2</sup> preserved these differences (from 25±10 to 31±10 mL/m<sup>2</sup>;  $P<0.001$ ), whereas indexing to BSA eliminated the gradient across BMI strata (from 41±17 to 42±14 mL/m<sup>2</sup>,  $P=0.232$ ; Table 2). Pre-A and minimum LA volumes showed a pattern similar to that of maximum LA volume, ranging from 51±32 to 72±31 mL and from 45±33 to 61±26 mL, respectively (both  $P<0.001$ ).

### LA Pressure, Function, and PV Relationships

LA maximum, minimum, mean, and pulse pressures were progressively higher across BMI groups (ranging from 17±7 to 25±11 mmHg, 8±5 to 13±6 mmHg, and 15±5 to 21±9 mmHg, respectively; all  $P<0.001$ ; Figure S1B; Table 2), whereas estimated LA stiffness remained comparable between groups (median 0.5–0.6 mmHg/mL;  $P=0.782$ ). The proportion of patients with elevated LA pressure (mean LA pressure ≥15 mmHg) was greater across BMI categories (Figure 1A; Table 2). Accordingly, the LA PV relationship showed an ordered rightward shift across BMI strata (Jonckheere-Terpstra  $P<0.001$ ; Figure 1B). The magnitude of LA wall stress oscillation throughout the cardiac cycle was larger across BMI strata (from 122±64 to 196±91 kdynes/cm<sup>2</sup>;  $P<0.001$ ; Figure 1C), predominantly driven by higher maximal wall stress (from 143±85 to 226±108 kdynes/cm<sup>2</sup>;  $P<0.001$ ; Figure 1D; Table 2). Mean LA pressure was also incrementally higher across CHA<sub>2</sub>DS<sub>2</sub>-VASC categories, ranging from ≈9 mmHg in patients with a score of 0 to ≈14 mmHg in those with scores ≥5 ( $P<0.001$ ; Figure S2). To account for potential confounding, we performed additional analyses adjusted for AF substrate surrogates, namely AF type (paroxysmal or nonparoxysmal) and prior/repeated AF ablation (none or ≥1). These analyses showed that BMI remained significantly associated with a load-related pattern of LA remodeling, characterized by larger LA volume, higher mean LA pressure, and greater cyclic wall stress (Table S3).

Phasic LA function, assessed by 2-dimensional echocardiography, showed weak overall associations with BMI. In contrast, among patients with available speckle-tracking analysis, higher BMI was correlated with lower absolute phasic LA strain values (|LA reservoir strain|:  $r=-0.32$ ,  $P<0.001$ ; |LA conduit strain|:  $r=-0.26$ ,  $P=0.008$ ; |LA contractile strain|:  $r=-0.18$ ,  $P=0.063$ ; Figure 2). Consistently, complementary markers such as the LA function index also showed lower values across higher BMI categories, both when indexed to BSA (29±17 to 20±13;  $P=0.005$ ) and when indexed to height<sup>2</sup> (48±27 to 27±17;  $P<0.001$ ; Table 2). Finally, estimated blood and plasma volume

**Table 2. Baseline Left Atrium Characteristics According to Body Mass Index Strata**

|                                                | BMI <25.0 kg/m <sup>2</sup><br>(N=179) | BMI 25.0–29.9 kg/m <sup>2</sup><br>(N=439) | BMI 30.0–34.9 kg/m <sup>2</sup><br>(N=298) | BMI ≥35 kg/m <sup>2</sup><br>(N=124) | P value |
|------------------------------------------------|----------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------|---------|
| LAV <sub>max</sub> , mL                        | 76±31                                  | 85±30                                      | 87±23                                      | 94±29                                | <0.001  |
| LAV <sub>pre-A*</sub> , mL*                    | 52±31                                  | 60±31                                      | 68±27                                      | 74±31                                | <0.001  |
| LAV <sub>pre-A*</sub> , MI                     | 51±32                                  | 58±25                                      | 67±27                                      | 72±31                                | <0.001  |
| LAV <sub>min</sub> , mL                        | 45±33                                  | 52±30                                      | 55±26                                      | 61±26                                | <0.001  |
| LAVi (BSA), mL/m <sup>2</sup>                  | 41±17                                  | 42±14                                      | 40±10                                      | 42±14                                | 0.232   |
| LAVi (height <sup>2</sup> ), mL/m <sup>2</sup> | 25±10                                  | 27±9                                       | 28±7                                       | 31±10                                | <0.001  |
| Total LAEF, %                                  | 41±17                                  | 40±16                                      | 40±16                                      | 38±14                                | 0.569   |
| Active LAEF, %                                 | 26±16                                  | 27±13                                      | 29±16                                      | 27±12                                | 0.409   |
| Passive LAEF, %                                | 27±14                                  | 26±12                                      | 25±13                                      | 24±14                                | 0.578   |
| LAFi (BSA)                                     | 29±17                                  | 24±15                                      | 22±13                                      | 20±13                                | 0.005   |
| LAFi (height <sup>2</sup> )                    | 48±27                                  | 37±23                                      | 32±19                                      | 27±17                                | <0.001  |
| Reservoir strain ( LASr )†                     | 24±12                                  | 17±9                                       | 15±8                                       | 12±6                                 | 0.005   |
| Conduit strain ( LAScd )†                      | 13±6                                   | 11±5                                       | 9±5                                        | 8±4                                  | 0.041   |
| Contractile strain ( LASct )†                  | 10 (5–17)                              | 6 (0–12)                                   | 3 (0–10)                                   | 2 (0–10)                             | 0.017   |
| LA stroke volume, mL                           | 28±14                                  | 31±14                                      | 35±17                                      | 37±18                                | <0.001  |
| LA maximum pressure, mmHg                      | 17±7                                   | 20±9                                       | 21±9                                       | 25±11                                | <0.001  |
| LA minimum pressure, mmHg                      | 2 (0; 4)                               | 2 (0; 5)                                   | 2 (0; 5)                                   | 2 (0; 7)                             | 0.010   |
| LA mean pressure, mmHg                         | 8±5                                    | 10±5                                       | 11±5                                       | 13±6                                 | <0.001  |
| LA mean pressure ≥15 mmHg, n (%)               | 13 (7)                                 | 64 (15)                                    | 56 (19)                                    | 37 (30)                              | <0.001  |
| LA pulse pressure, mmHg                        | 15±5                                   | 17±8                                       | 18±8                                       | 21±9                                 | <0.001  |
| LA stiffness, mmHg/mL                          | 0.5 (0.4–0.8)                          | 0.5 (0.3–0.8)                              | 0.5 (0.3–0.8)                              | 0.6 (0.4–0.8)                        | 0.782   |
| LA wall stress max, kdynes/cm <sup>2</sup>     | 143±85                                 | 166±83                                     | 185±81                                     | 226±108                              | <0.001  |
| LA wall stress min, kdynes/cm <sup>2</sup>     | 13 (0–36)                              | 16 (0–36)                                  | 19 (0–37)                                  | 19 (0–45)                            | 0.154   |
| LA wall stress delta, kdynes/cm <sup>2</sup>   | 122±64                                 | 144±68                                     | 161±68                                     | 196±91                               | <0.001  |

BMI indicates body mass index; BSA, body surface area; LA, left atrial; LAEF, left atrial emptying fraction; LAFi, left atrial function index; LAV, left atrial volume; and LAVi, left atrial volume index.

\*Available only in patients in sinus rhythm.

†Available in 107 patients (10%). LA strain components are reported as absolute values.

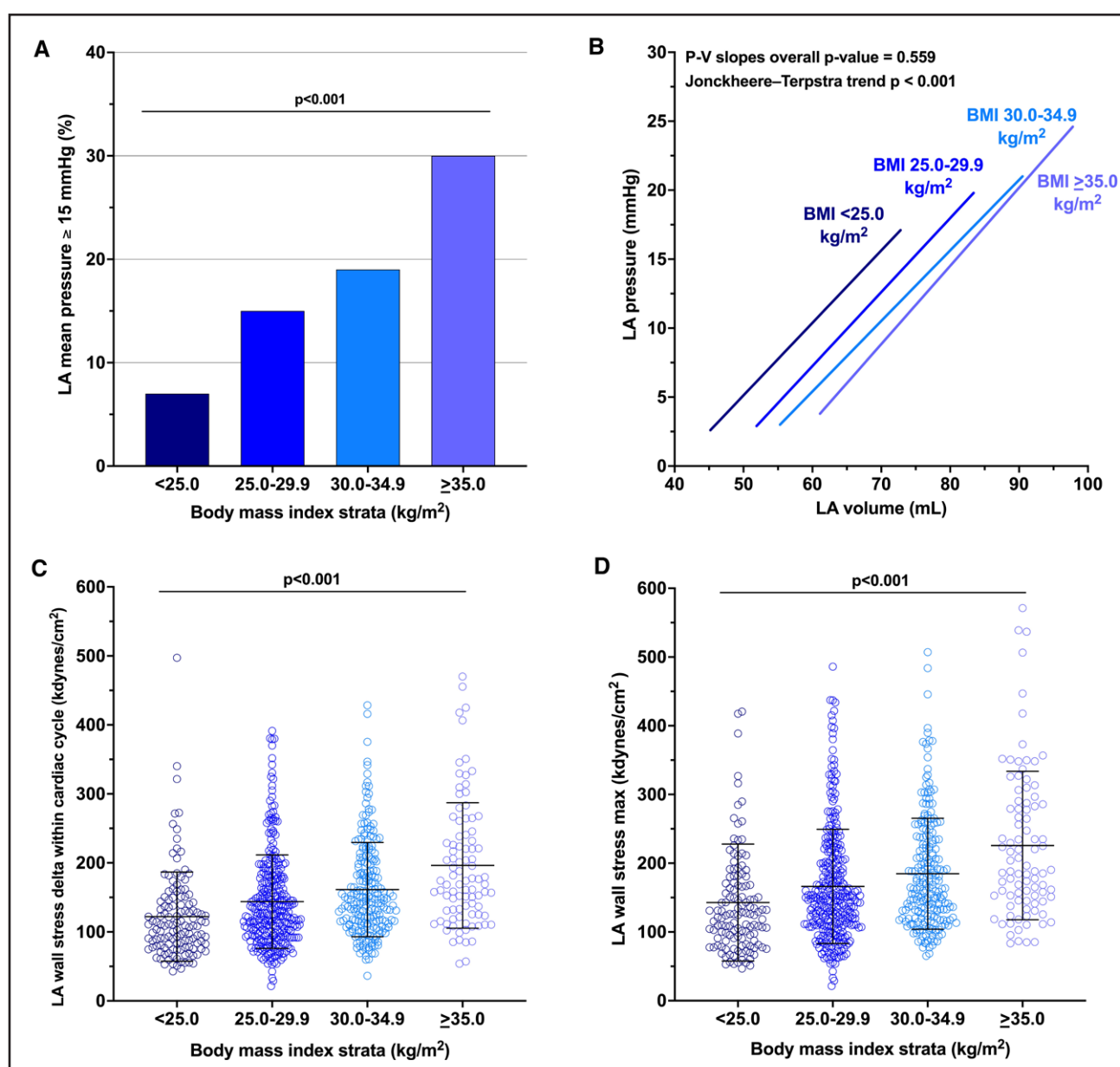
were positively correlated with LA volume ( $r=0.26$  and  $r=0.29$ , respectively; both  $P<0.001$ ; Figure 3A; Figure S3). Larger LA volume was also correlated with a consistent pattern of phasic functional impairment, with lower values across all strain components (|LA reservoir strain|:  $r=-0.43$ ; |LA conduit strain|:  $r=-0.32$ ; |LA contractile strain|:  $r=-0.36$ ; all  $P<0.001$ ; Figure 3B).

Among study participants, those with HF or LVEF  $\leq 40\%$  ( $n=171$ ) had larger LA volumes (99 versus 87 mL;  $P<0.001$ ) and more impaired LA functional indices, including lower total LAEF (33% versus 41%;  $P=0.001$ ) and lower active LAEF (20% versus 28%;  $P<0.001$ ). As expected, LA pressures were substantially higher in the HF subgroup, which translated into increased LA wall stress (166±81 versus 206±108 kdynes/cm<sup>2</sup>;  $P<0.001$ ) and a greater wall stress delta (146±68 versus 172±86 kdynes/cm<sup>2</sup>;  $P=0.003$ ; Table S4). However, the BMI-related pattern of LA remodeling was preserved after exclusion of patients with HF and LVEF  $\leq 40\%$ , with materially unchanged findings in the restricted population (Table S5; Figure S4).

## DISCUSSION

In this large cohort of patients referred for catheter ablation of AF, higher BMI was associated with a pattern of LA remodeling predominantly occurring in parallel with greater estimated intravascular volume. This phenotype was characterized by larger LA size, higher LA pressures, a rightward shift of the estimated LA PV relationships, and increased cyclic wall stress.

Obesity is a well-established risk factor for AF and HFpEF,<sup>2,27</sup> conditions in which atrial remodeling plays a central pathophysiological role.<sup>23,28</sup> In this study, absolute LA volumes were incrementally larger across BMI categories, with a similar pattern observed for maximal, pre-A, and minimal LA volumes. In parallel, higher BMI was associated with greater estimated blood and plasma volumes that correlated directly with LA volume, suggesting that chronic intravascular volume expansion may contribute to LA enlargement in obesity.<sup>29</sup> This higher volume loading may promote passive atrial dilatation and electrical instability,<sup>30</sup> potentially preceding the development of overt phasic functional dysfunction.



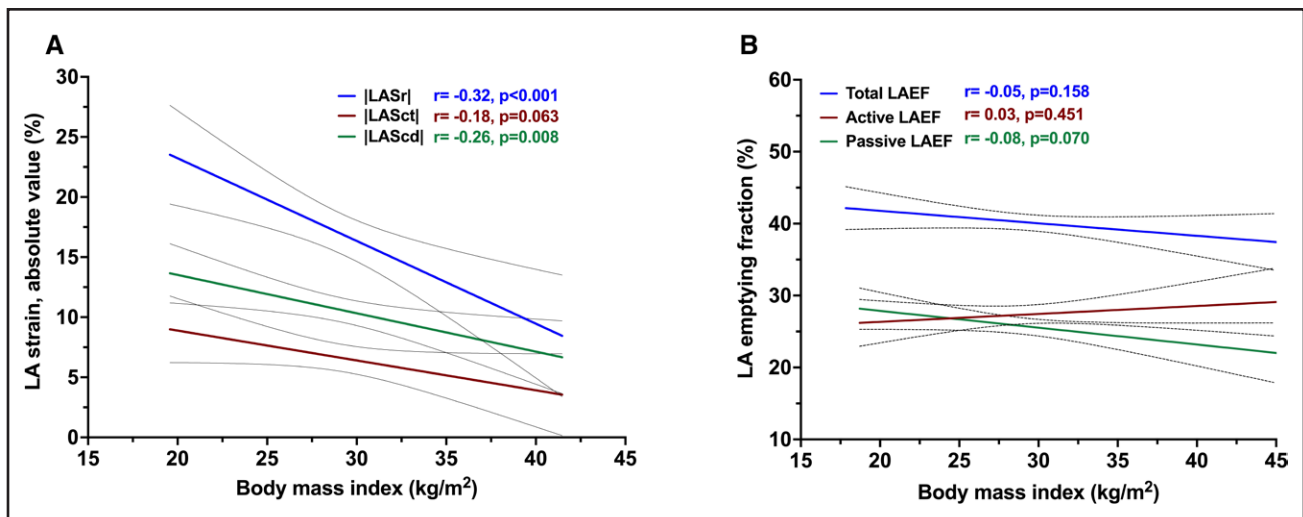
**Figure 1.** Left atrial (LA) pressure-volume and wall stress measures across body mass index (BMI) strata.

**A**, Shows the prevalence of elevated mean LA pressure across BMI strata. **B**, Shows the slope of the LA pressure-volume relationship across BMI strata. **C**, Shows the change in LA wall stress over the cardiac cycle. **D**, Shows maximum LA wall stress across BMI strata.

While intriguing, this interpretation should be considered hypothesis-generating, as the formulas used to estimate plasma and total blood volume are highly dependent on body weight. Notably, while the absence of lower hematocrit across higher BMI strata argues against isolated hemodilution, it does not exclude circulating volume expansion, as plasma volume and red cell mass may increase in parallel in obesity, leaving hematocrit unchanged or even slightly higher. Consistent with this interpretation, prior studies using direct blood-volume assessment have shown higher blood and plasma volumes together with higher total hemoglobin mass in individuals with overweight or obesity,<sup>31</sup> while

large population-based data also support an association between obesity and increased erythropoietic activity.<sup>32</sup> Beyond circulating volume, other mechanisms may also contribute to the observed load-related LA phenotype, including higher LV filling pressures, diastolic dysfunction,<sup>33</sup> and obesity-related changes in transmural LA/LV pressures due to increased extracardiac constraint.<sup>34</sup> These mechanisms are not mutually exclusive and may act in combination to promote LA enlargement, higher measured LA pressures, and greater cyclic wall stress.

Notably, this gradient was markedly attenuated after LA volumes were indexed to BSA. This pattern is consistent with evidence that isometric BSA indexing may



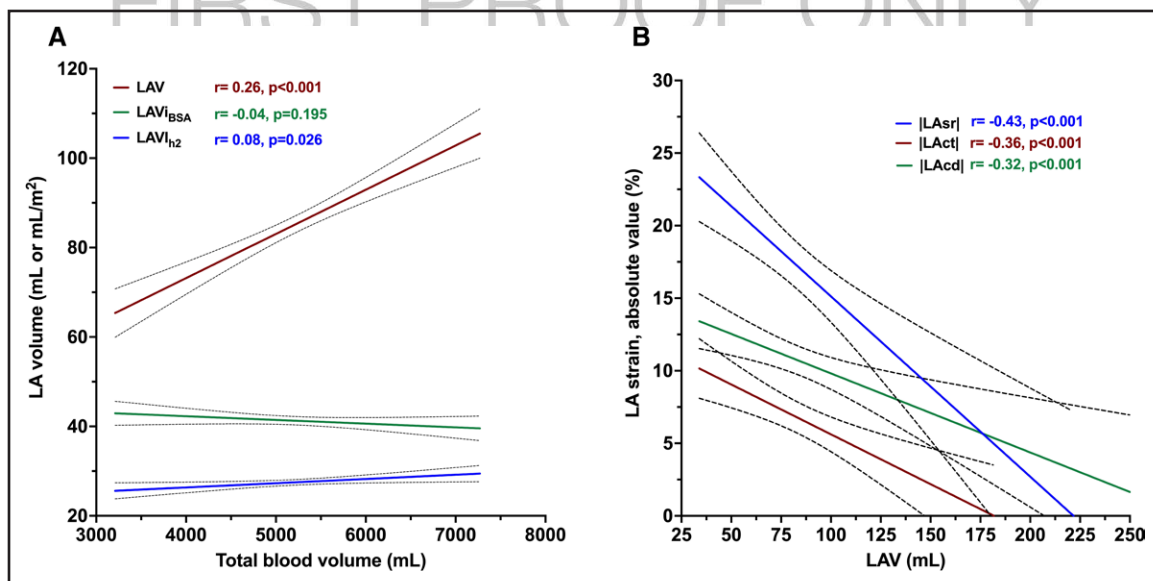
**Figure 2. Associations of body mass index with left atrial (LA) functional measures.**

**A**, Shows the associations between body mass index and LA strain components, reported as absolute values. **B**, Shows the associations between body mass index and 2-dimensional echocardiographic measures of LA function. LAEF indicates left atrial emptying fraction; LAScd, left atrial conduit strain; LASct, left atrial contractile strain; and LASr, left atrial reservoir strain.

overcorrect atrial size in obesity, because organ growth is not strictly linear with body size and because BSA is disproportionately influenced by fat mass in obese individuals.<sup>35</sup> In contrast, allometric approaches, particularly height-based scaling, seem to better preserve proportionality between indexed and absolute LA size and may reduce the risk of underrecognition of pathological atrial enlargement in obesity.<sup>36</sup> More recently, the LA end-systolic to LV end-diastolic volume ratio has been proposed as an alternative to body-size indexing in patients with obesity, as it may better reflect atrioventricular

proportionality and has been shown to be more closely associated with diastolic dysfunction, LV filling pressures, and obesity-related HFpEF than indexed LA volume alone.<sup>37,38</sup>

Consistent with a predominantly load-related phenotype, LA systolic, diastolic, mean, and pulse pressures were incrementally higher across BMI strata, whereas estimated LA operant stiffness and the slope of the LA PV relationship remained largely preserved across BMI strata. This dissociation may suggest that obesity-related atrial remodeling at this stage is characterized



**Figure 3. Associations of blood volume and left atrial volume (LAV) with left atrial (LA) structure and function.**

**A**, Shows the associations between estimated total blood volume and LAV, LAV indexed (LAVi) to body surface area (BSA), and LAVi to height squared. **B**, Shows the associations between LA strain components and LAV. On the y-axis, volumes are expressed as mL for LAV, mL/m<sup>2</sup> for LAVi indexed to BSA, and mL/m<sup>2</sup> for LAVi indexed to height squared. LAScd indicates left atrial conduit strain; LASct, left atrial contractile strain; and LASr, left atrial reservoir strain.

by an adaptive rightward shift of the LA PV relationship, reflecting volumetric accommodation rather than clearly demonstrable intrinsic myocardial stiffening. Such hemodynamic behavior is distinct from the marked increases in atrial stiffness and reduced compliance observed in established atrial cardiomyopathy seen in more advanced HF stages<sup>23,39</sup> and is consistent with contemporary models describing atrial disease as a continuum ranging from early adaptive enlargement to overt atrial myopathy and atrial failure.<sup>40,41</sup> However, it should be noted that the PV parameters used in our study were derived from simplified 2-point estimates rather than from continuous, time-resolved PV loops. Therefore, estimated LA stiffness and the interpretation of PV shifts should be viewed with caution, and subtle changes in intrinsic atrial material properties across BMI strata cannot be excluded.

Despite no significant difference in estimated LA stiffness, higher BMI was associated with a marked increase in LA wall stress, particularly with greater variation across the cardiac cycle. This observation is mechanistically interesting, as cyclic atrial stretch and dynamic wall-stress fluctuations have been shown to promote electrophysiological instability through stretch-activated signaling pathways, enhanced mechano-electric feedback, and altered calcium handling (ie, by mechanisms that trigger and further facilitate atrial arrhythmogenesis).<sup>42</sup> Both experimental models and clinical studies further demonstrate that atrial regions exposed to higher wall stress are preferentially associated with fibrosis, low-voltage substrate, and conduction abnormalities.<sup>43</sup> The emphasis on increased wall-stress variability, rather than early intrinsic stiffening or contractile dysfunction, provides a mechanistic signal supporting the hemodynamic interpretation of the findings.

Interestingly, when LA function was assessed using conventional volumetric indices, only weak associations with BMI were observed, whereas strain-derived indices were lower across BMI categories in parallel with increasing LA size. In this context, LA strain may provide supportive information on early BMI-associated atrial functional abnormalities, capturing subtle alterations that precede overt LA dysfunction.<sup>44</sup> As a hypothesis-generating observation, the lower LA reservoir strain observed at higher BMI may also be compatible with early diastolic dysfunction,<sup>45</sup> in which subclinical functional atrial abnormalities emerge before overt LA disease, increased atrial stiffness, or the clinical manifestation of HF.<sup>44,46</sup> However, because strain data were available only in a limited subset, these findings should be considered exploratory. Similarly, the LA function index, a prognostic composite measure integrating volumetric and flow-related components,<sup>47</sup> also showed lower values at higher BMI, potentially suggesting the presence of a load-dependent atrial functional impairment.

Sensitivity analyses in patients with HF or reduced LVEF provide additional context for this mechanistic framework. Indeed, this cohort exhibited larger LA volumes, more impaired volumetric atrial function (particularly reduced active LA emptying fraction), higher atrial pressures, and markedly increased LA wall stress and wall stress variation compared with patients without HF. Notably, mean LA pressure was higher across increasing CHA<sub>2</sub>DS<sub>2</sub>-VASc categories, indicating an association between global clinical risk burden and atrial pressure load. Collectively, these findings are compatible with a more advanced atrial myopathy phenotype characterized by chronic volume overload, reduced atrial contractile reserve, and maladaptive functional remodeling,<sup>23</sup> distinct from the predominantly load-related pattern observed across higher BMI in the overall cohort.

While the present findings are compatible with a predominantly load-related pattern of LA remodeling in obesity, it is important to acknowledge that obesity may promote atrial arrhythmogenesis through additional, nonhemodynamic mechanisms that were not directly assessed in this study. Previous data indicate that excess adiposity is associated with lipid accumulation within the atrial myocardium, epicardial adipose tissue expansion, and myocardial lipotoxicity, processes that may directly impair atrial electrophysiological properties and structural integrity.<sup>48–50</sup> Moreover, obesity-related insulin resistance and metabolic dysregulation have been linked to mitochondrial dysfunction, oxidative stress, and altered calcium handling within atrial cardiomyocytes, all of which may further facilitate atrial arrhythmogenesis independently of chamber loading conditions.<sup>51,52</sup> These mechanisms are likely to act in parallel with hemodynamic stress and may become increasingly relevant at more advanced stages of obesity-related atrial disease.

From a clinical standpoint, these findings raise the possibility that BMI-associated LA remodeling may represent an early and potentially modifiable contributor to arrhythmogenesis and HF pathophysiology in individuals with obesity. With increasing adiposity, atrial enlargement is accompanied by higher filling pressures, resulting in amplified LA wall stress and greater oscillations in wall tension due to the parallel increase in both chamber volume and pressure.<sup>53</sup> Interventions aimed at sustained weight reduction and optimization of volume status may therefore represent plausible strategies to mitigate LA remodeling and potentially slow progression toward advanced atrial dysfunction.<sup>53,54</sup> While assessment and treatment of clinically evident congestion may represent a biologically plausible strategy to reduce atrial loading conditions, there is currently no evidence that this approach improves LA dysfunction or geometry in obese patients with AF and LA enlargement. Beyond fluid management, a holistic risk-factor management approach may improve AF-related outcomes. Consistently, the recent ARREST-AF randomized trial showed that structured lifestyle and

risk-factor modification improved freedom from recurrent atrial arrhythmia after AF ablation in patients with elevated BMI and cardiometabolic risk factors, together with reductions in body weight, waist circumference, and systolic blood pressure.<sup>55</sup> Prior longitudinal studies in patients with obesity undergoing bariatric surgery have demonstrated significant improvements in LA reservoir strain and volumes, suggesting that obesity-related atrial dysfunction is at least partially reversible and primarily driven by loading and metabolic factors rather than fixed atrial myopathy.<sup>56,57</sup> More recently, pharmacological weight-loss strategies, particularly glucagon-like peptide-1 receptor agonists, have been shown to reduce the incidence of AF in individuals with overweight or obesity.<sup>58</sup> Treatment with semaglutide for 1 year significantly reduced LA volume as compared with placebo in the STEP-HFpEF program, providing further support for the relationship between fat mass and LA remodeling.<sup>59</sup> These benefits may extend beyond weight loss alone and plausibly involve favorable effects on atrial loading conditions, systemic inflammation, epicardial adiposity, and myocardial remodeling.<sup>60</sup> While our data cannot fully unravel this complex scenario, the evidence discussed above suggests the hypothesis that weight-reduction strategies may help attenuate LA remodeling in patients with obesity.

### Strengths and Limitations

This study includes notable strengths, including the simultaneous measurement of both LA pressure and volume in over 1000 patients, which is unprecedented in the literature, enabling comprehensive characterization of LA structure and function.

Our findings should also be considered in the context of the following limitations. First, the cross-sectional study design precludes causal inference. The study population is a selected cohort of patients referred for AF ablation at a single tertiary center, potentially introducing referral and selection biases. Patients were relatively young and predominantly in an early stage of atrial myopathy, with a low prevalence of HF and long-standing persistent AF. Therefore, these findings may not be directly generalizable to older patients or to more advanced populations with extensive LA structural and functional damage. Moreover, because all patients had AF and were referred for catheter ablation, concomitant AF-related atrial disease may have influenced the observed associations. Although additional analyses accounting for AF substrate surrogates showed consistent findings, residual confounding (ie, AF burden, disease duration) cannot be excluded. HF status was based on a preexisting clinical diagnosis documented in medical records and LVEF  $\leq 40\%$ , rather than systematic adjudication. Accordingly, HF phenotype could not be reliably classified, and residual confounding by HF and related comorbidities cannot

be excluded. Estimated plasma and total blood volumes were derived from anthropometric and hematologic formulas rather than being directly measured. As a consequence, the volume/load-related interpretation should be considered hypothesis-generating, and residual confounding cannot be excluded. Notably, prior AF catheter ablation may be associated with increased LA pressure and stiffness, likely reflecting procedure-related atrial scarring and reduced compliance, and may therefore have influenced the hemodynamic and functional measurements in the present study.<sup>61</sup> Invasive LA PV relationships were derived from discrete minimum and maximum pressure and volume measurements, without continuous temporal alignment of pressure and volume signals across the cardiac cycle. Accordingly, estimated operant LA stiffness and PV slopes reflect simplified 2-point estimates rather than fully time-resolved PV loops. LV end-diastolic volumes were not systematically measured using biplane volumetric methods, preventing robust calculation of the LA end-systolic to LV end-diastolic volume ratio and limiting assessment of atrioventricular proportionality across BMI strata. Data on the prevalence and severity of obstructive sleep apnea were not available. Given its established association with AF and atrial remodeling, it may represent an additional mechanistic pathway contributing to atrial disease. However, obstructive sleep apnea is also closely linked to intravascular volume expansion and peripheral edema, supporting the concept that hemodynamic load-related mechanisms may still play a central role within the proposed framework. LA strain analysis was available only in a subset of patients due to practical and technical factors, potentially limiting statistical power and generalizability. Moreover, while patients with and without strain data were broadly comparable at baseline, unmeasured differences and selection bias cannot be excluded. Therefore, strain findings should be interpreted as exploratory and hypothesis-generating. Similar caution applies to derived measures such as LA emptying fractions and LA function index, as these indices combine multiple volumetric and flow-related measurements and may therefore be particularly vulnerable to cumulative measurement error. We could not reliably assess the prognostic value of LA structural, functional, or hemodynamic measures for HF- or AF-related outcomes, as follow-up clinical care was frequently managed in local hospitals outside our institution. Finally, the absence of a control group without AF represents a further limitation. However, the risks associated with transseptal puncture in individuals without a clinical indication outweigh potential research benefits.

### Conclusions

In patients with a history of AF, higher BMI was associated with a distinct pattern of LA remodeling occurring in parallel with greater estimated chronic volume load. This

phenotype was characterized by LA enlargement, elevated atrial pressures, a rightward shift in the estimated LA PV relationship, and greater cyclic wall stress, without significant differences in estimated operant LA stiffness. However, given the simplified approach used for PV estimation, subtle changes in intrinsic atrial material properties may have been missed, thereby limiting the interpretation of estimated LA stiffness and PV parameters. Overall, these findings suggest the potential existence of an early, load-related pattern of BMI-associated atrial remodeling, preceding overt atrial dysfunction, and potentially contributing to the link between excess adiposity, AF, and HFpEF.

## ARTICLE INFORMATION

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### Supplemental Material

Tables S1–S5

Figures S1–S4

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